Lecture 03: Entity-Relationship Diagram (Part 1 Core Theory)

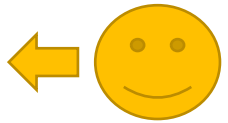
ITCS216 Fundamentals of Database Systems
Semester 1/2026

Asst. Prof. Dr. Preecha Tangworakitthaworn preecha.tan@mahidol.ac.th
Asst. Prof. Dr. Jidapa Kraisangka jidapa.kra@mahidol.ac.th
Asst. Prof. Dr. Siripen Pongpaichet siripen.pon@mahidol.ac.th

This lecture is adopted from ITCS241 content in 1/2022 and collarborate with Asst. Prof. Dr. Preecha Tangworakitthaworn

Database Topics

Knowing How to Develop Data Model for Business Application

- Entity-Relationship Model
- **Developing the ER Diagram** 
- Developing a Relational Database Schema from an ER Diagram

Lesson Learning Outcomes

At the end of this lecture, the students will be able to:

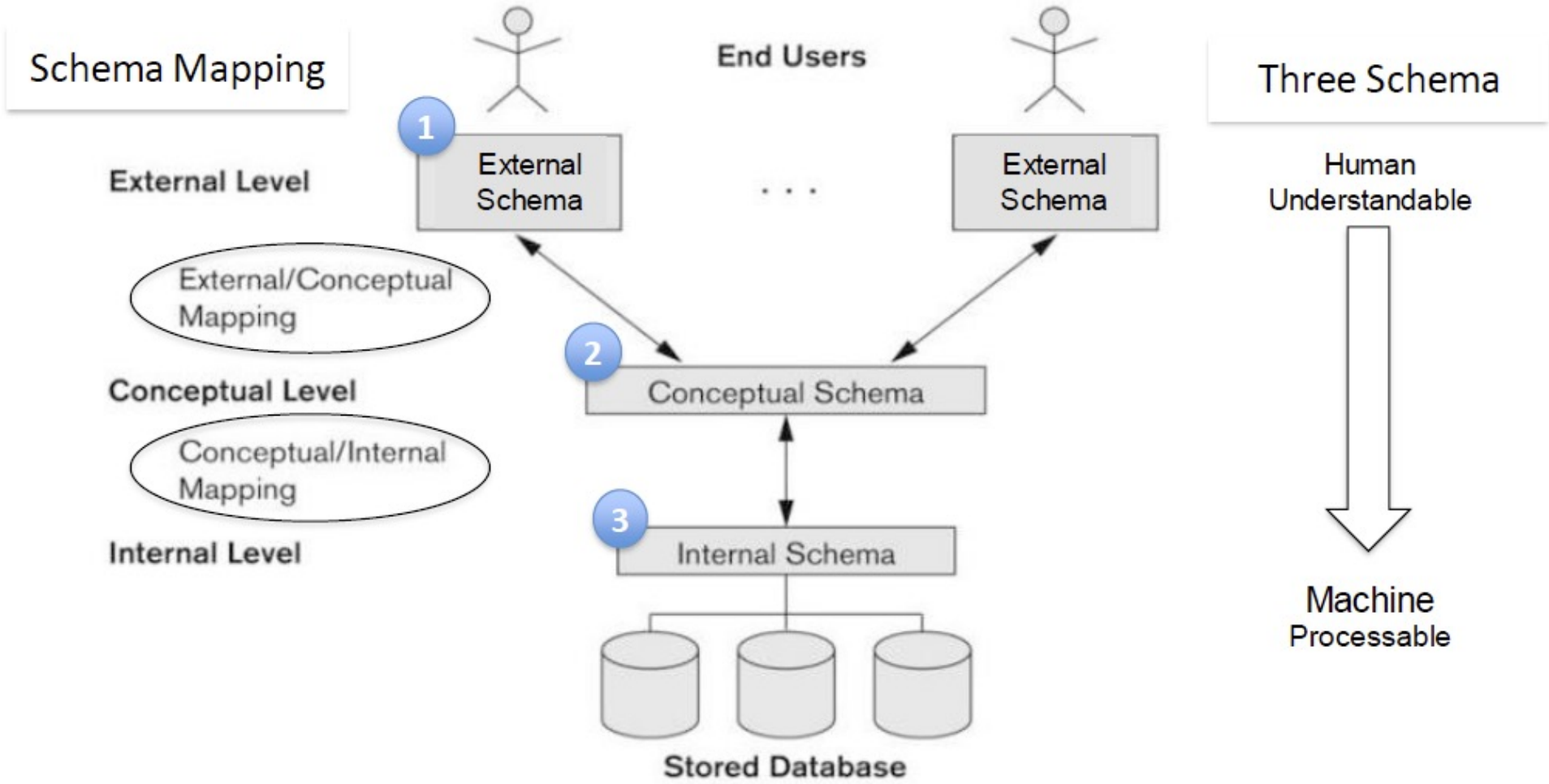
- Know the concept and method to develop ERD.
- Understand the important relationship patterns.
- Understand the representation of business rules in an ER diagram.
- Know the symbols and vocabulary of the Chen's and Crow's Foot notation for ER diagram.
- Use the connectivity symbols to represent 1-1, 1-M, and M-N relationships.

RECAP

Database Concept

Recommend

Three Schema Architecture



RECAP

Business Rule's Template

Recommend

- **Template #1**

Data Object Noun	Relationship Verb	Data Object Noun

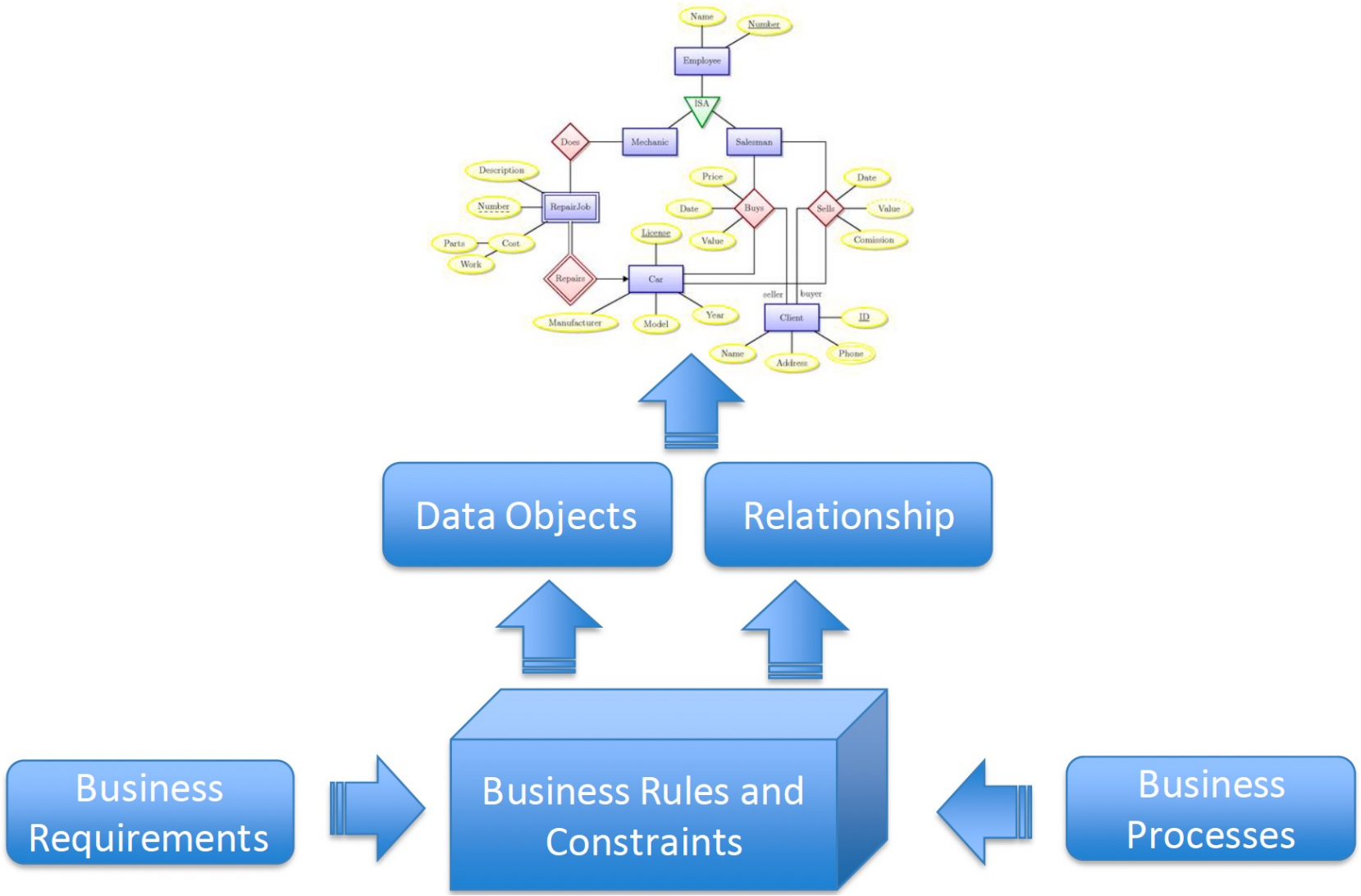
For example: A Student attends many Classes.

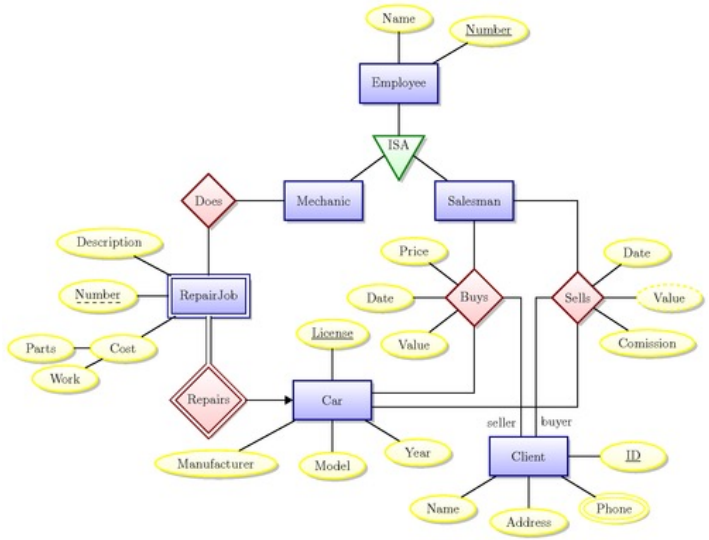
- **Template #2**

Data Object Noun	Container Verb	Data Properties/Attributes Noun

For example: A Student comprises Student ID, Name, and many EMails.

Business Rules as building box for data model creation

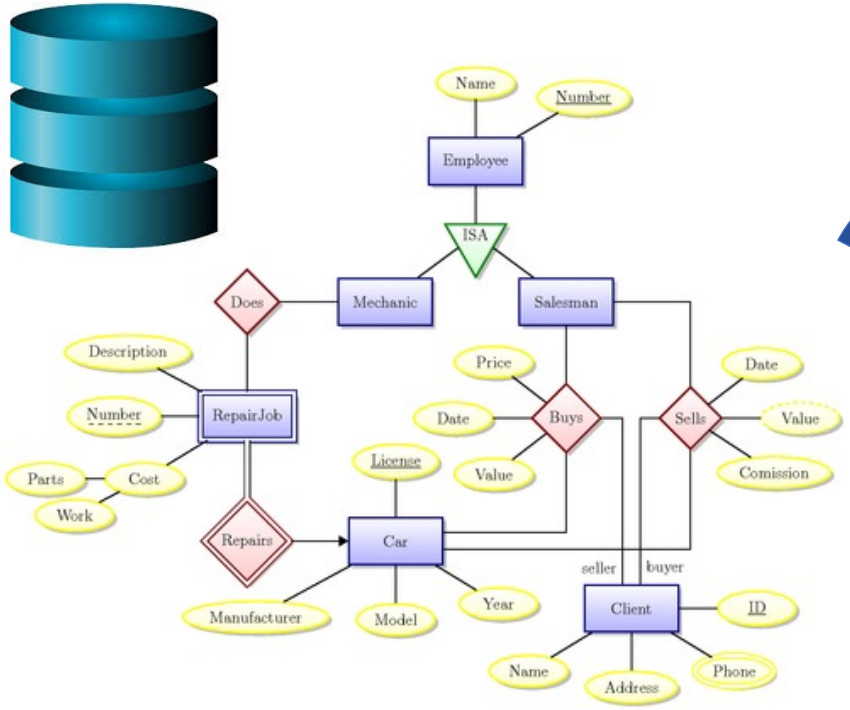




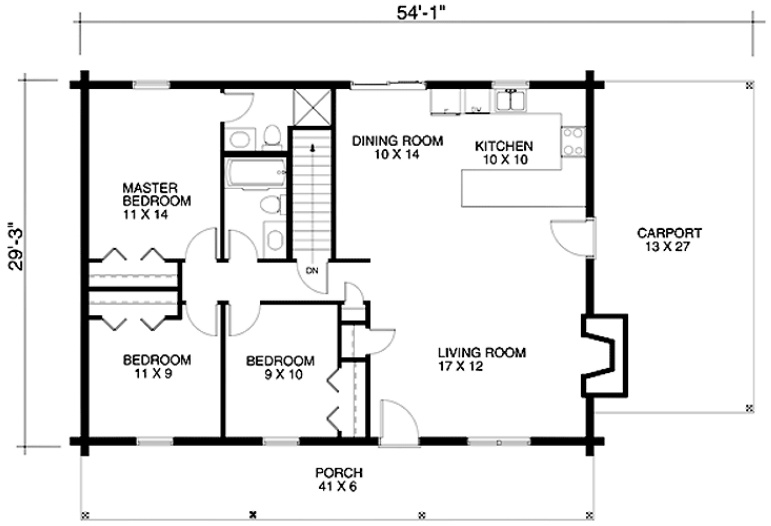
Part 1
Introduction to ERD

ERD: Entity Relationship Diagram

DATA MODEL = BLUEPRINT







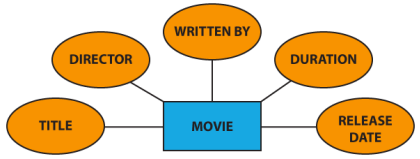
ERD: Entity Relationship Diagram
Two well-known ER notations

Chen Notation

Peter Chen

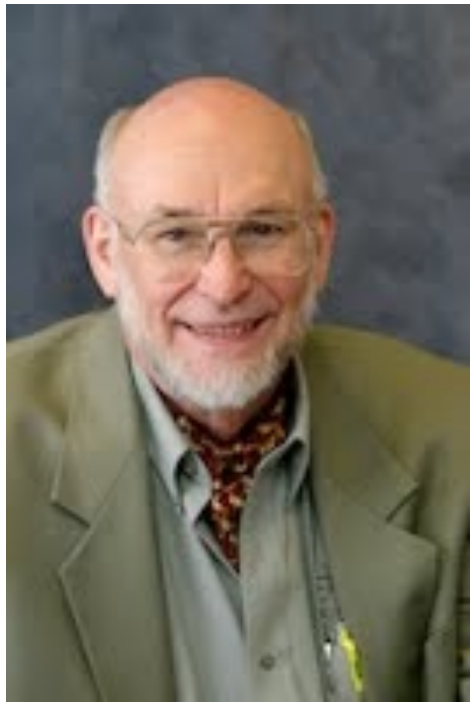


http://www.csc.lsu.edu/~chen/



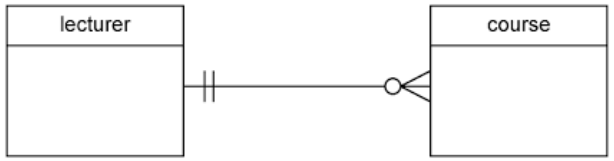
Crow's Foot Notation

Gordon Everest

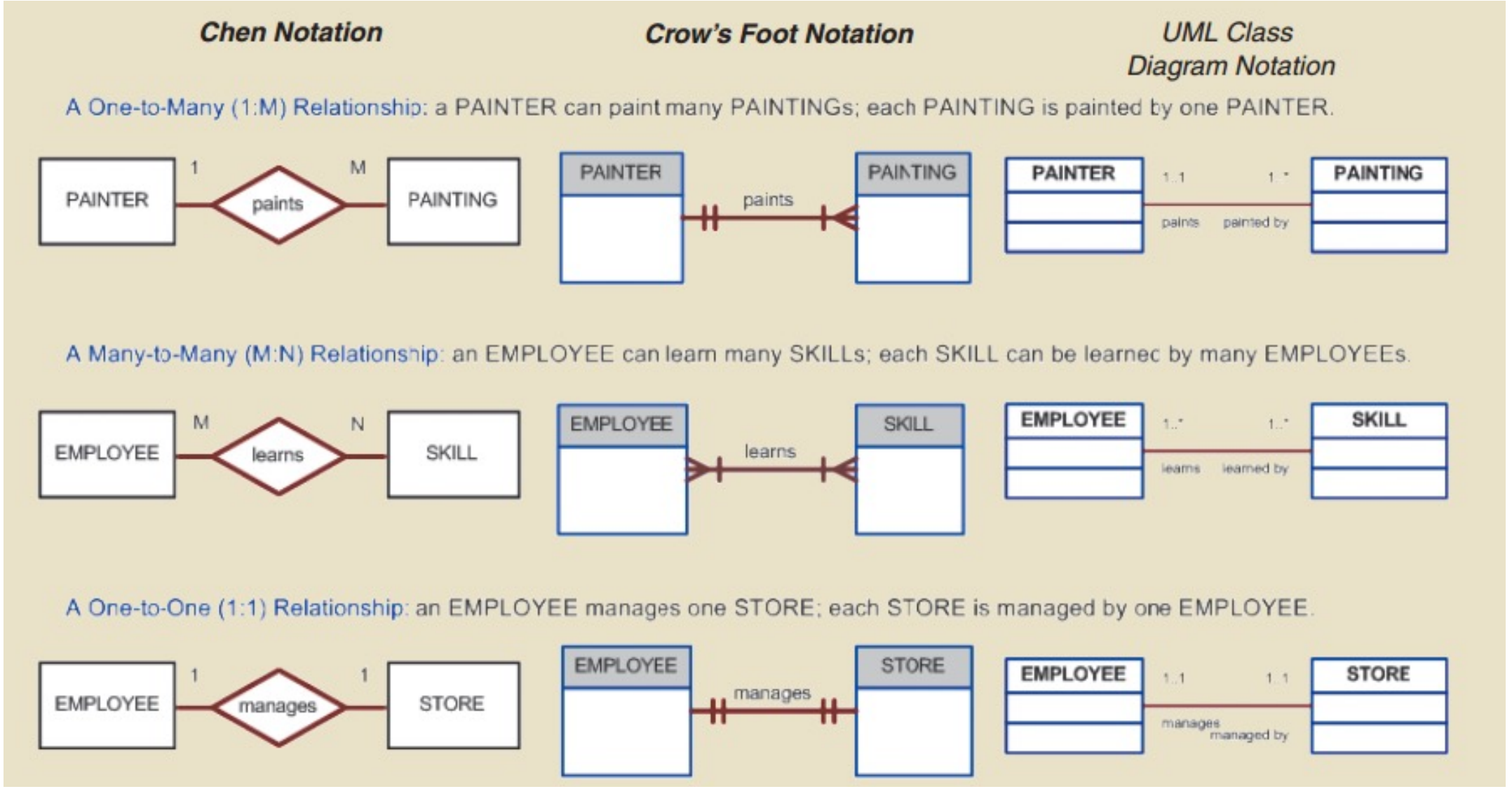




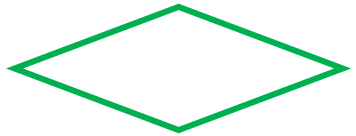
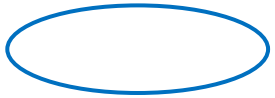
http://geverest.umn.edu/



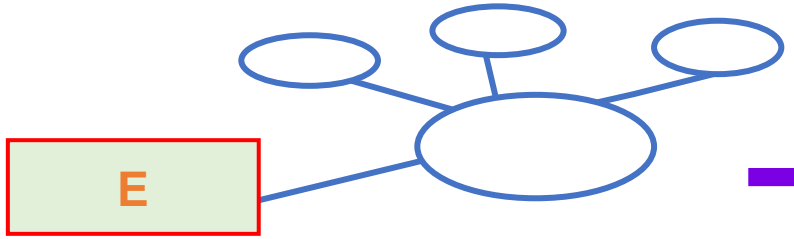
ERD Notations



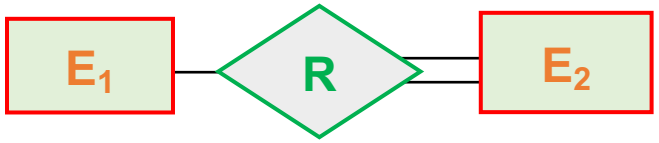
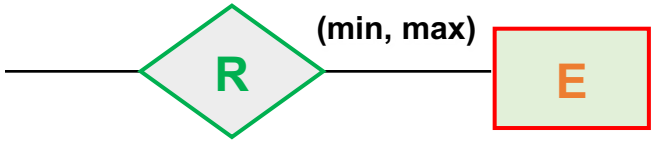
Chen’s Notations (1)

Symbol	Meaning
	**Entity** type
	Weak entity type
	**Relationship** type
	**Attribute**

Chen's Notations (2)

Symbol	Meaning
	**Attribute**
	Key attribute (Identifier/Primary key)
	Multivalued attribute
	Composite attribute
	Derived attribute

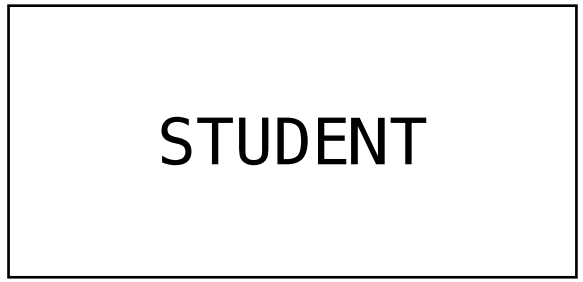
Chen's Notations (3)

Symbol	***Meaning***
	Total participation of E_2 in R
	Connectivity constraint 1:N for E_1 : E_2 in R
	Cardinality constraint (min, max) on participation of E in R

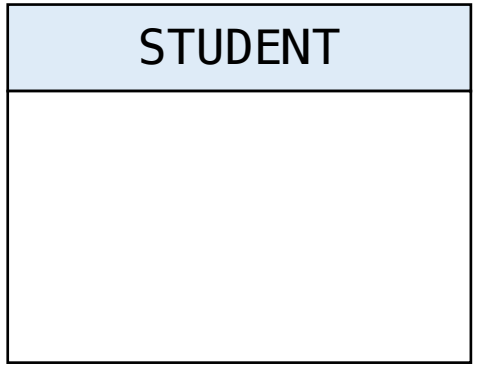
Entities

- A **data object** representing meaningful raw data.
- Refer to **entity set.**
- A table in relational data model.
- A rectangle notation contains entity name.

Chen model



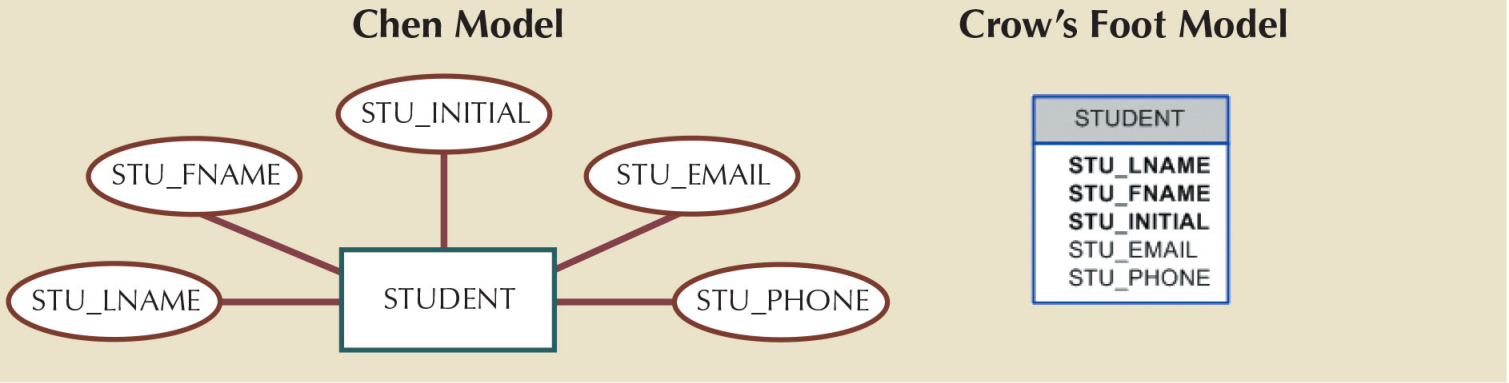
Crow's foot model



Attributes

- The properties of an entity.
- Such as, a student has first name, last name, initial, email, and telephone number.

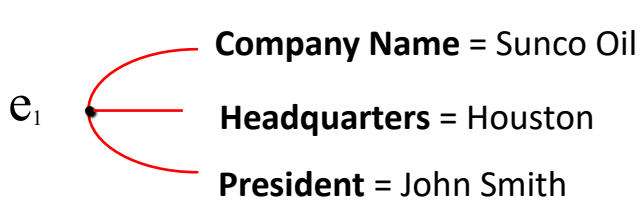
FIGURE 4.1 THE ATTRIBUTES OF THE STUDENT ENTITY: CHEN AND CROW'S FOOT

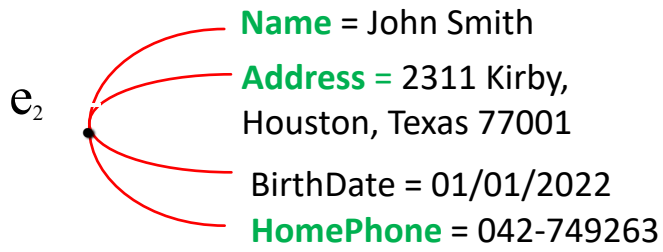


Types of Attributes

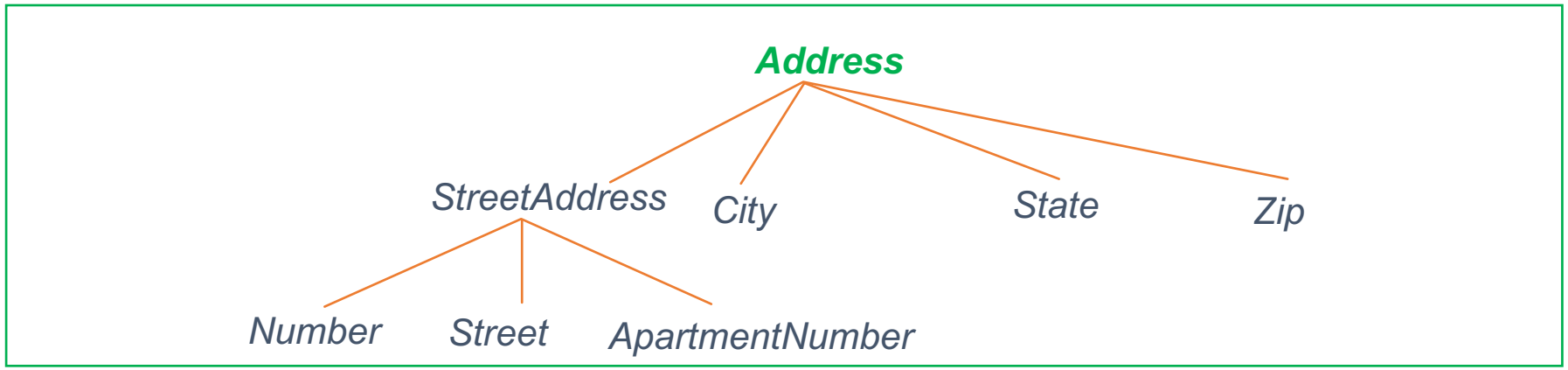
1. Simple vs. Composite







Example of Composite attributes



Types of Attributes – (2)

2. Single-valued vs. Multivalued



student ID, gender

color** of a car, college **degrees, address

Example of Multivalued attributes

Color of car:	*Red (Roof)*	*Black (Body)*	*Blue (Trim)*	*White (Interior)*
College Degree:	*B.SC.*	*M.Sc.*	*Ph.D.*	
Address:	*Current*	*Permanent*	*Work*	

Types of Attributes – (3)

So-called
Information

3. Stored vs. Derived (Computed Attributes)



Birthdate

Current age, Number of employees

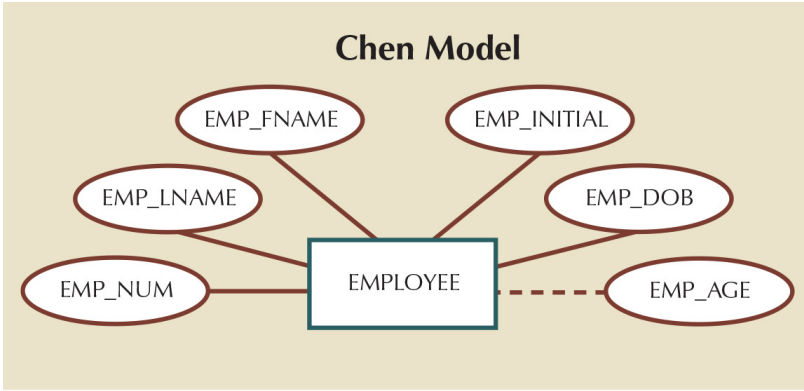


Example of derived attribute

Age = Year(Current Date) – Year(Birthdate)

of employees = COUNT(active employees)

Total Credits = SUM(enrolled subjects' credits)



Domain (or Data Type) of Attribute

- Attributes have a domain.
- Domain is a set of **possible/permitted values** of a given **attribute**.
  - **GENDER** IS CHARACTER (M, F) or Boolean (0,1)
  - **Score** IS NUMERIC (0, 100)
  - **ADDRESS** IS STRING (50)
- An attribute can have only a single domain.

Key Attribute of an Entity

- **Key attribute** (identifier/primary key) is attribute(s) whose values are **distinct** (or unique) for each entity.
  - For example, a student entity has a student's ID as a key attribute.

- This property is called **key constraint** or a **uniqueness constraint**.

- Key attribute may consist of
  - a single attribute.
  - a combination of attributes (or a composite key)
    - e.g., suppose no student will have the same firstname and lastname, then these two attributes can be used together to identify a particular (single) student.

Strong vs Weak Entity

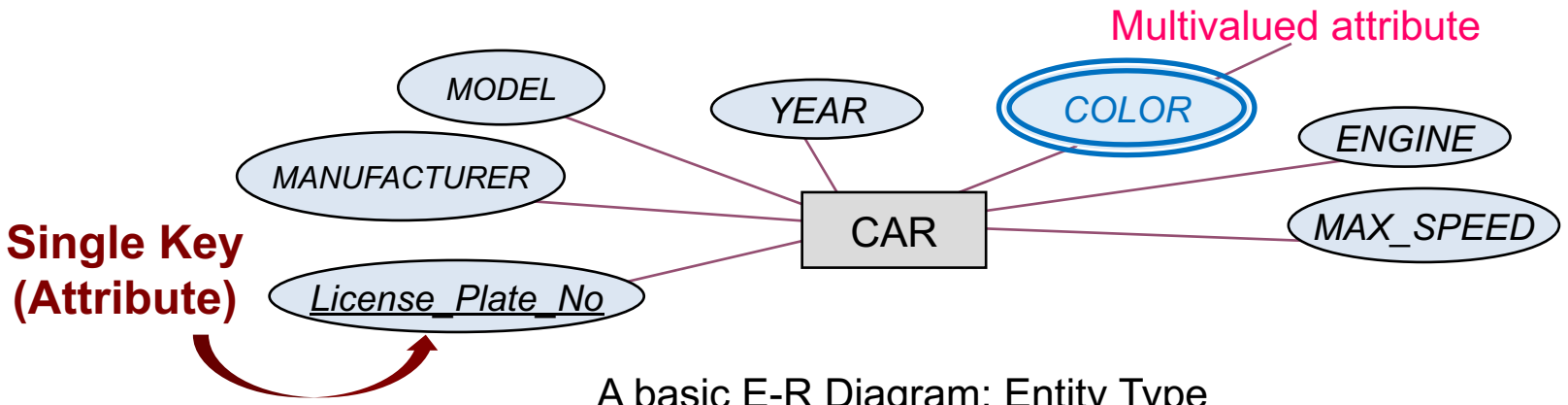
- The entity that represents the main data object of a specific business process is called the **strong entity.**

- The entity that meets two conditions is called a **weak entity**:

  1. *The entity is existence-dependent; it cannot exist without the strong entity with which it has a relationship, OR*
  2. *The entity has a primary key that is partially or totally derived from the parent entity (or strong entity) in the relationship.*

Key Attribute of an Entity

CAR (License_Plate_No, MANUFACTURER, MODEL, YEAR, {COLOR},
ENGINE, MAX_SPEED)



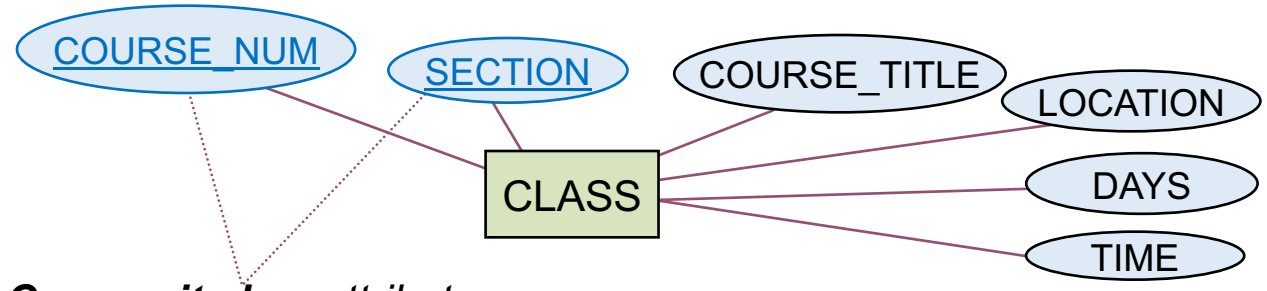
A basic E-R Diagram: Entity Type

License_Plate_No attribute *uniquely identifies* each entity instance.

Key Attribute of an Entity

CLASS (COURSE_NUM, SECTION, COURSE_TITLE, LOCATION, DAYS, TIME)

Composite Key (Attributes)



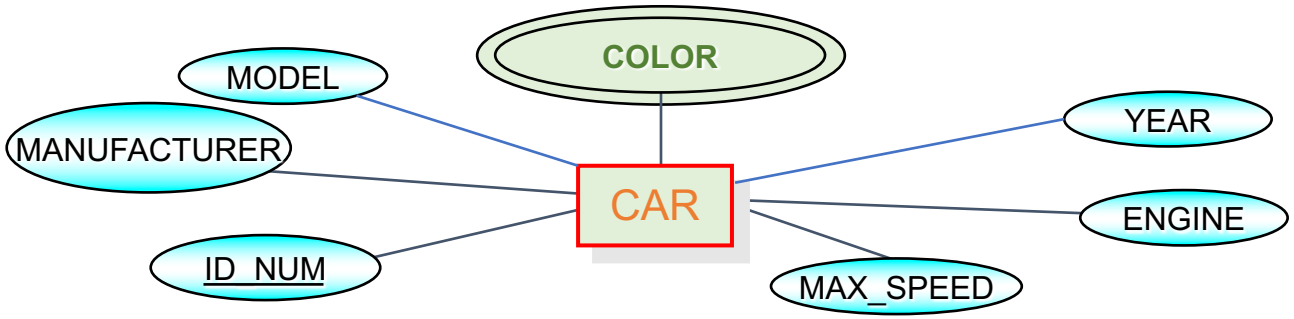
***Composite key** attributes*
(COURSE_NUM + SECTION are unique for each class)

ENTITY : CLASS

COURSE_NUM	SECTION	COURSE_TITLE	LOCATION	DAYS	TIME
CIS-420	1	Database Design and Application	KOM-202	W	6:00-8:40
CIS-420	2	Database Design and Application	PH-105B	MWF	9:00-9:50
CIS-220	1	Intro. to Microcomputers	KOM-204	TTh	11:00-12:15

Multivalued Attribute

COLOR → TOPCOLOR, BODYCOLOR, TRIMCOLOR

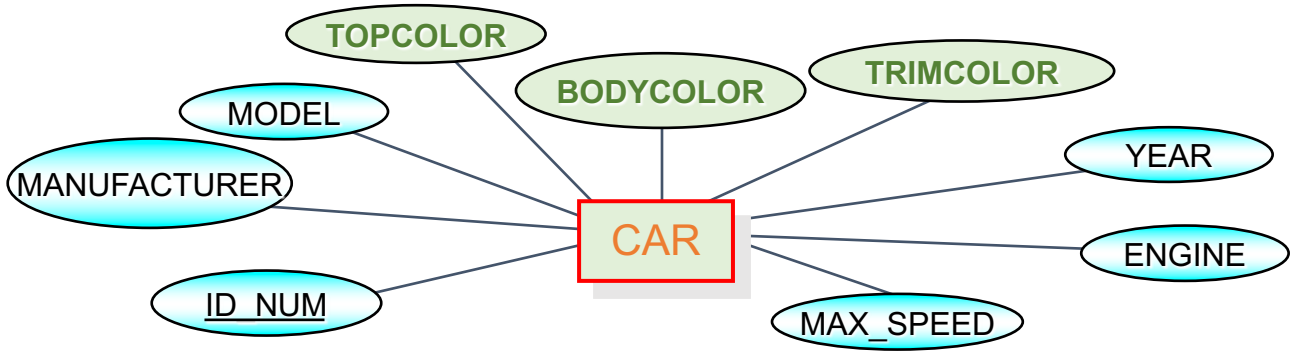


Multivalued attribute is not an ideal solution!! Why?

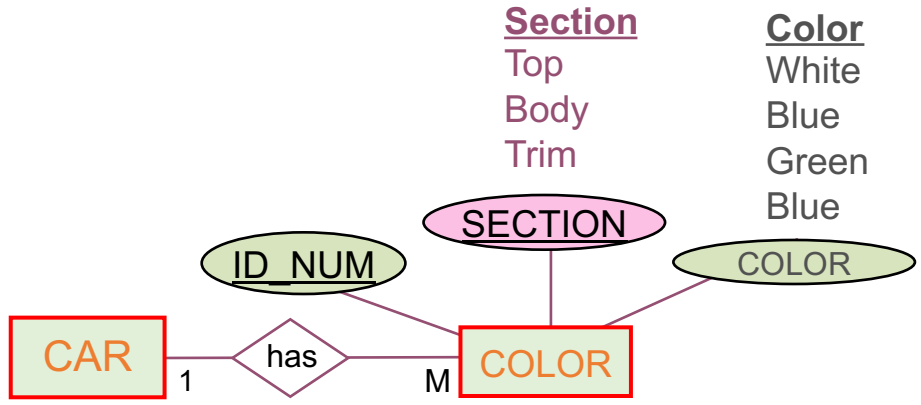
Remove Multivalued Attribute

Method1: Create several new attributes to represent the original attribute.

COLOR → TOPCOLOR, BODYCOLOR, TRIMCOLOR



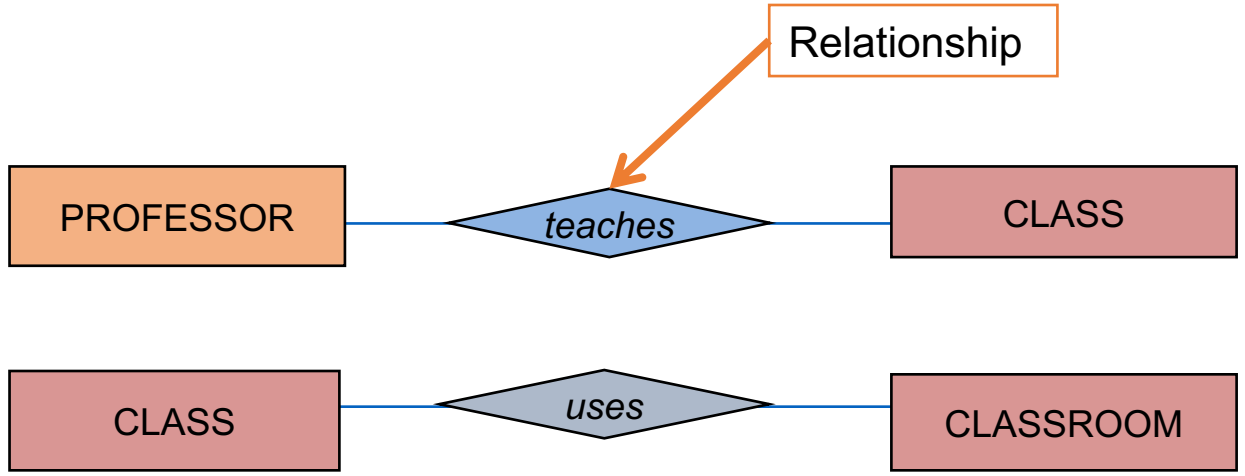
Method2: Create A NEW ENTITY to represent the original attribute with 1:M relationship.



A New Entity Set of a Multivalued Attribute's Components

Relationships

- Two entities must have relationship to each other.
- Relationship is an association among entities.



Three Types of Relationship

One-to-many (1:M or 1..*) relationship

- A painter paints many different paintings, but each one of them is painted by only one painter

Many-to-many (M:N or *..*) relationship

- An employee may learn many job skills, and each job skill may be learned by many employees.

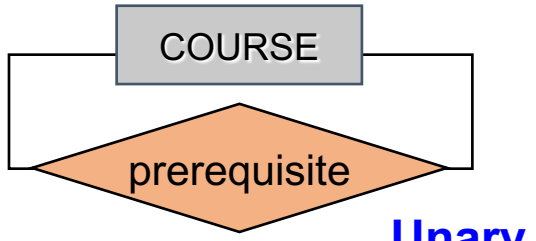
One-to-one (1:1 or 1..1) relationship

- Each of the stores be managed by a single employee.
- Each store manager, who is an employee, manages only a single store.

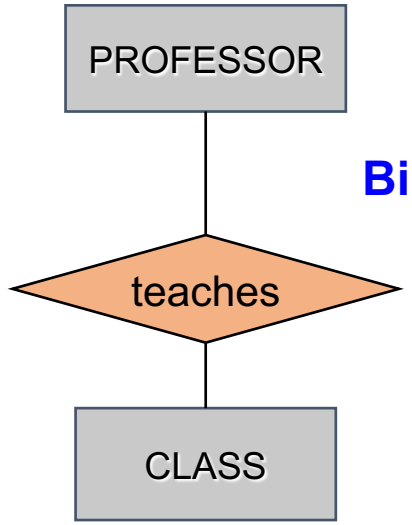
Note: Relationships are BIDIRECTIONAL ← →

Degrees of Relationship

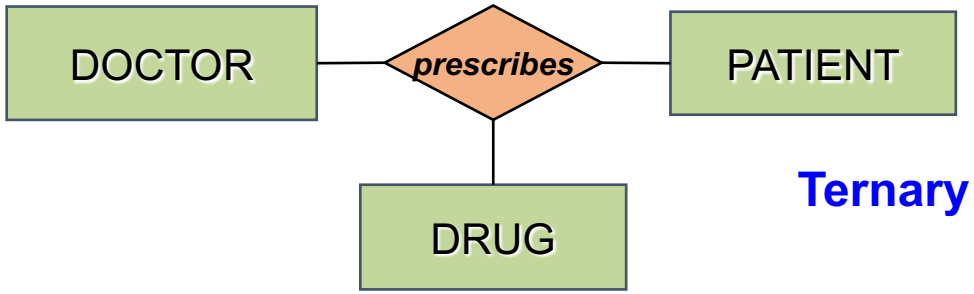
- A relationship degree indicates the number of entities associated with a relationship
  - Unary (1)
  - Binary (2)
  - Ternary (3)
  - N-ary (>3)



Unary



Binary

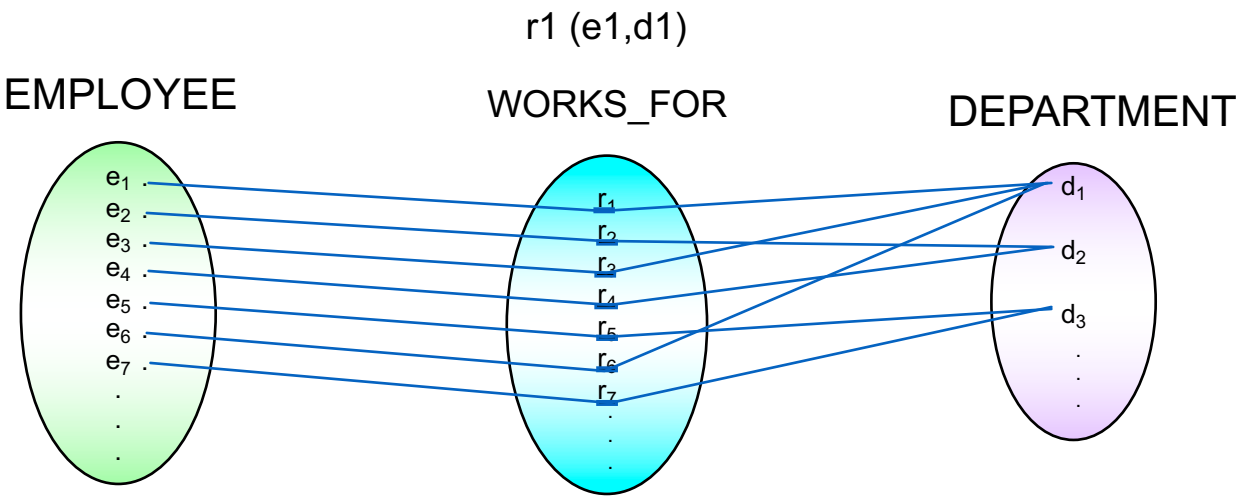


Ternary

Degree of a Relationship Type

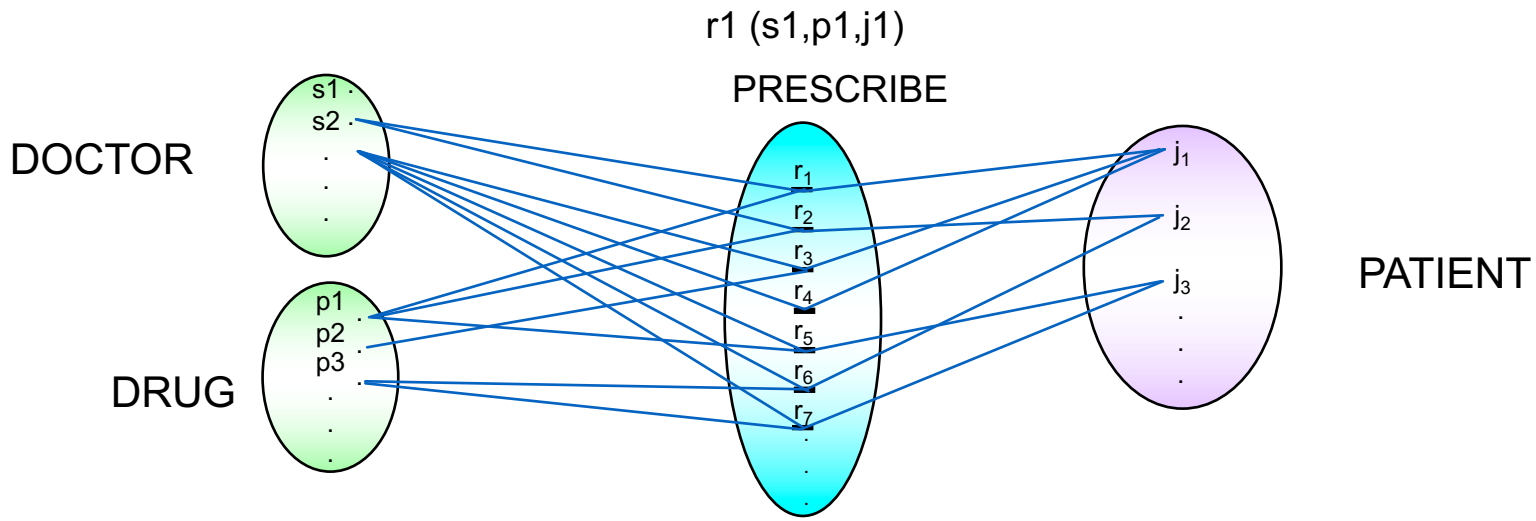
- The number of participating entities in that relationship type

Binary relationship



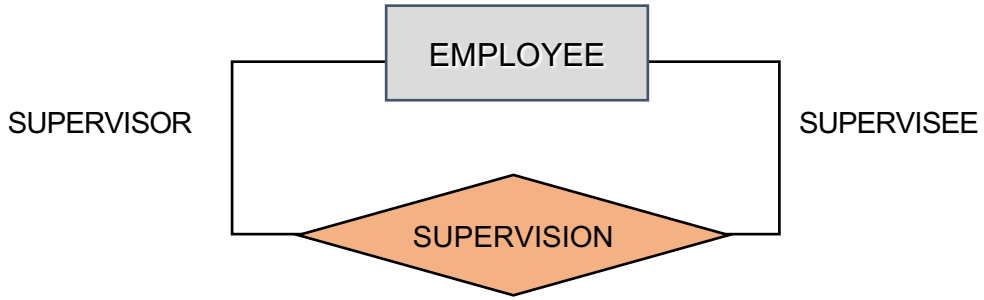
Degree of a Relationship Type

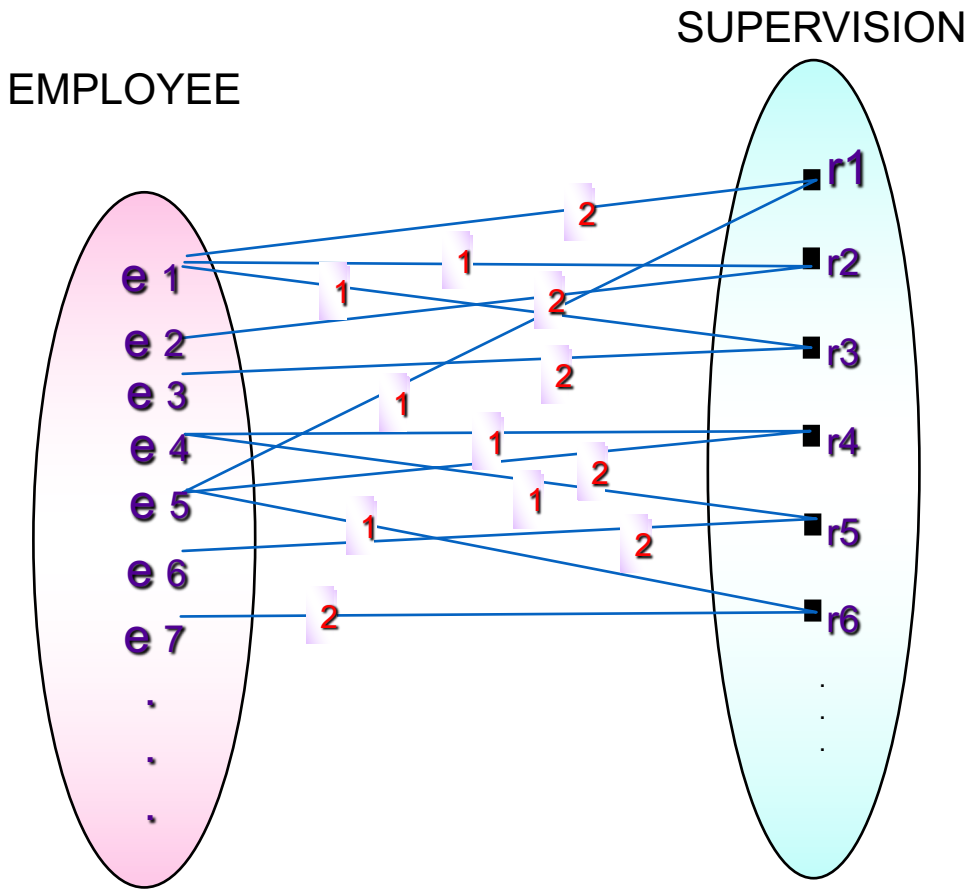
Ternary relationship

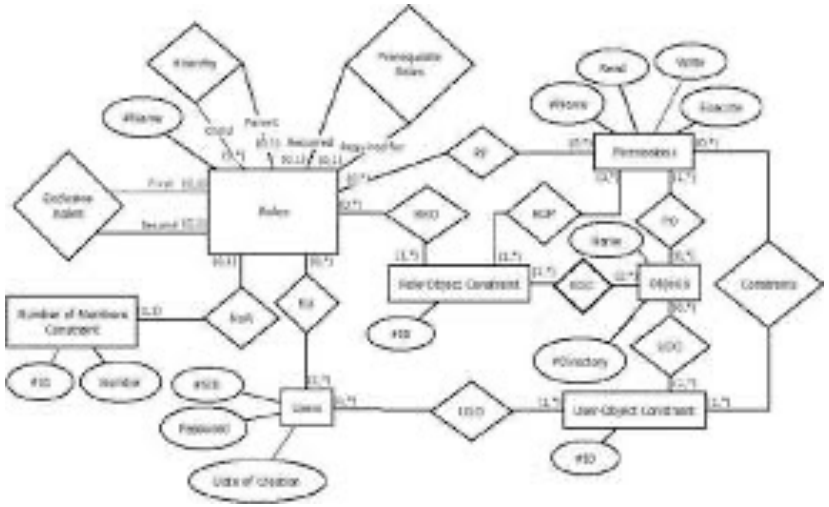


Recursive Relationship

- Same entity type participates in the relationships type more than once with the different roles.







Part 2
ERD Development in details

Mapping Constraints

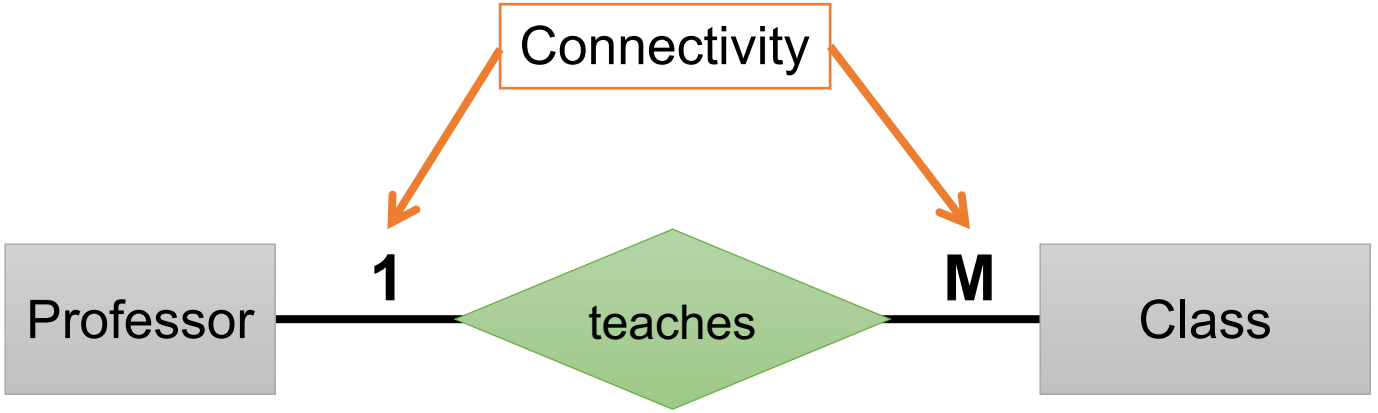
- Constraints are pre-defined rule which the content of a database must conform.

Relationship Constraints

- **Connectivity** Constraint
- **Cardinality** Constraint
- **Participation** Constraint (Total VS Partial)

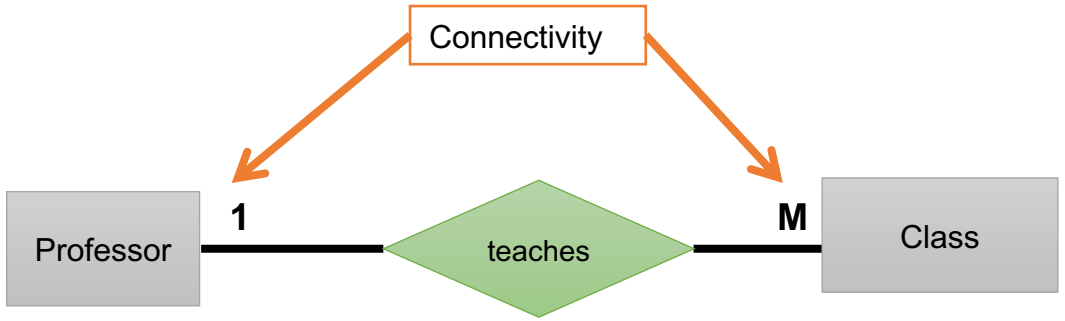
Connectivity Constraint

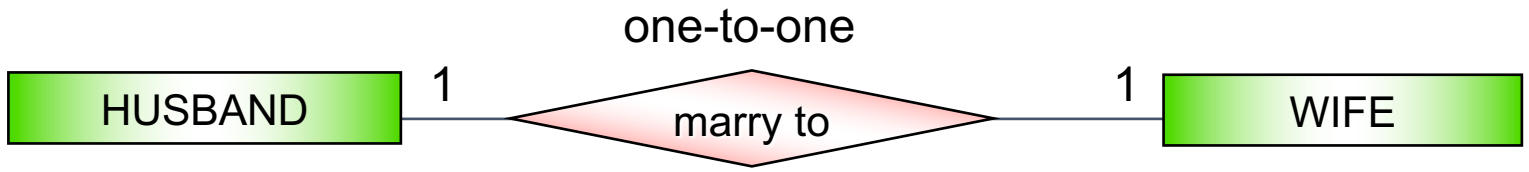
- **Connectivity**: used to describe the relationship classification. Just to show the relation existence and type of relation between entities.

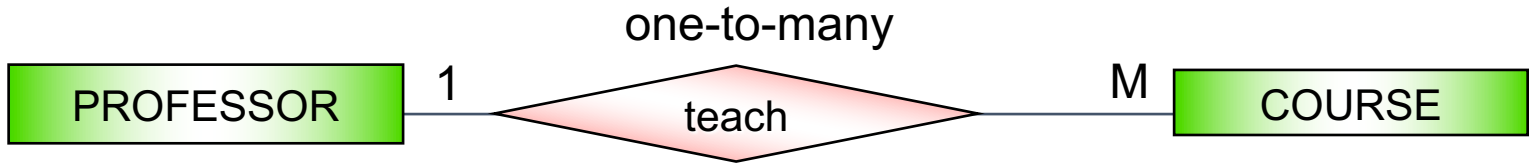


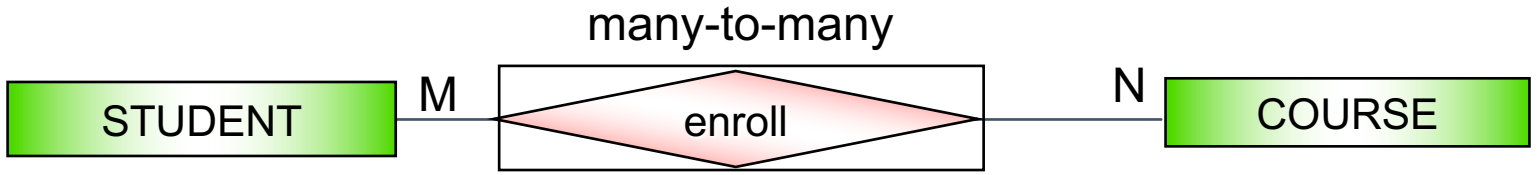
Connectivity Constraint

- One-to-One
- One-to-Many
- Many-to-One
- Many-to-Many





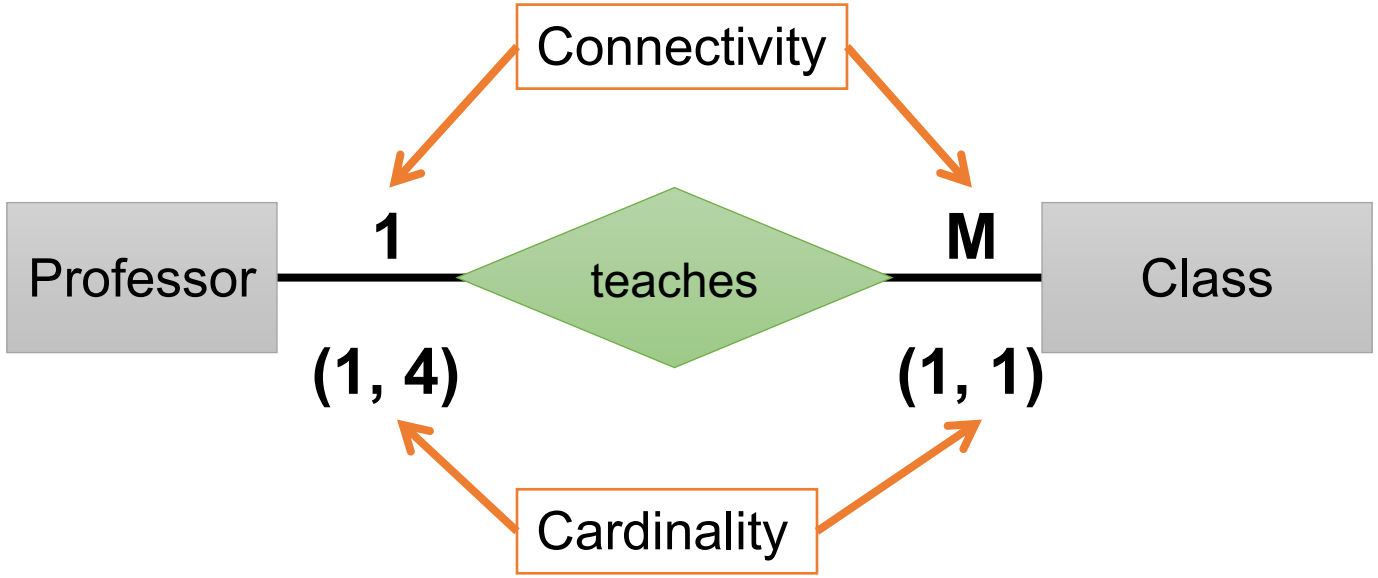




Cardinality Constraint

- **Cardinality** – a specific value assigned for the connectivity.
  - It expresses the range (minimum & maximum) of allowed entity occurrence (data records) associated with a single occurrence of the related entity.
  - Can be established from business rules/constraints
  - Can be written in the form of "(Min, Max)"
  - Example:
    - The maximum credit for a student to register is 21 credit per semester.
    - The minimum subject for a student to enroll is 5 subjects, while the maximum is 7 subjects.

Cardinality Constraint
(Chen Symbols)



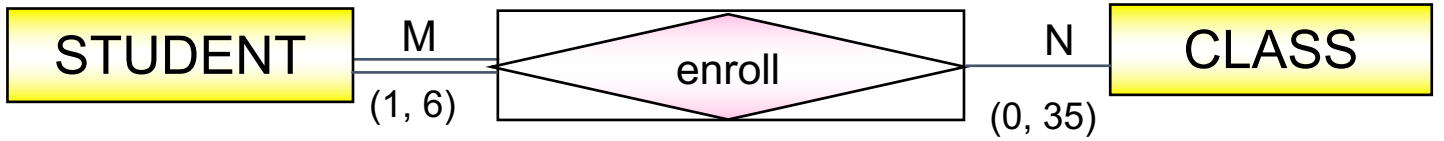
Cardinality Constraint (Chen Symbols)

1:M relationship



A professor can teach 0 up to 3 classes.
A class require at least 1 and at most 1 professor.

M:N relationship



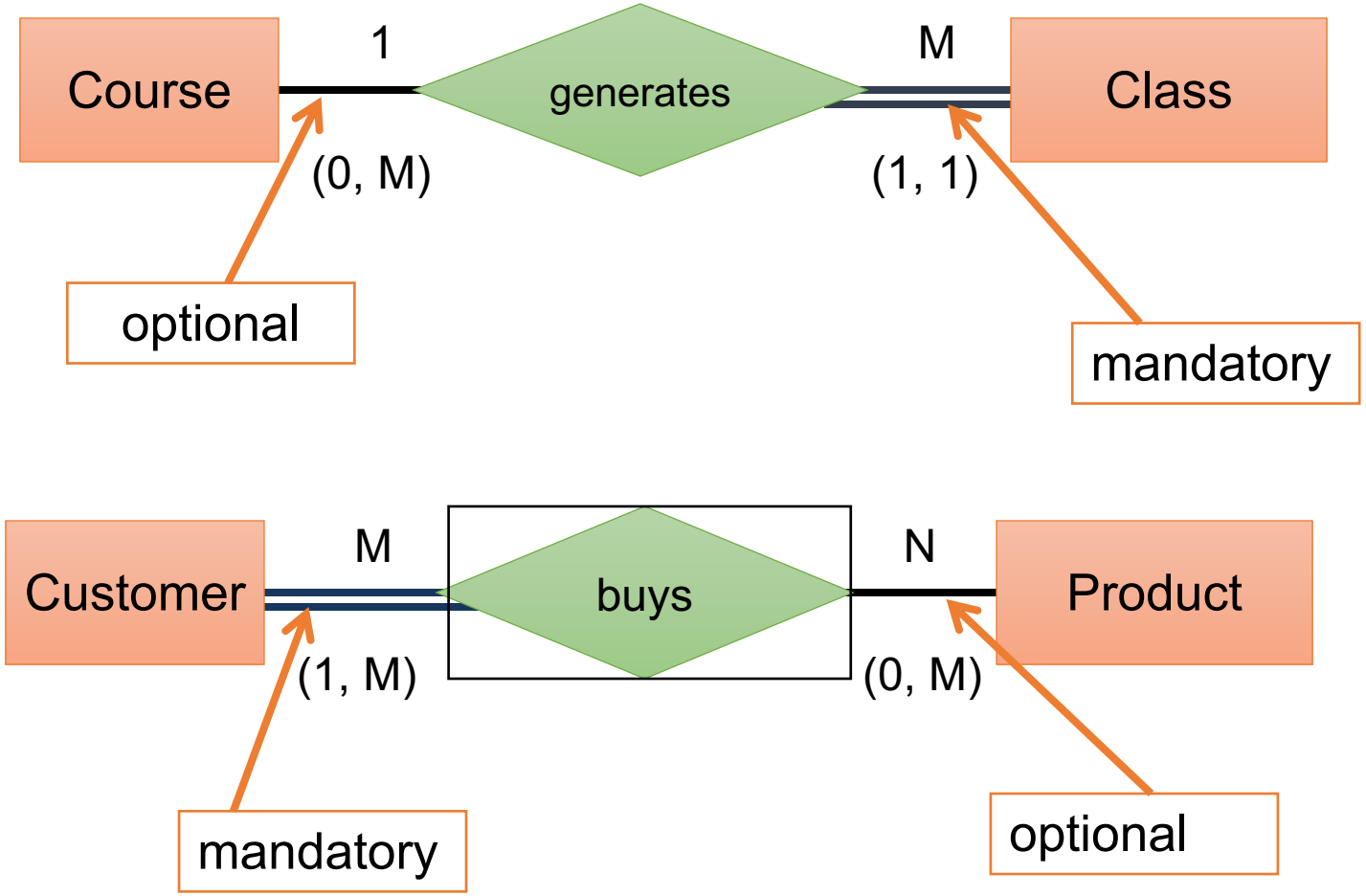
A student can enroll minimum of 1 class or maximum of 6 classes.
A class can have no student or 35 students at most.

Participation Constraint

- **Optional** participation (Partial Participation)
  - One entity occurrence does not require corresponding entity occurrence in particular relationship
  - In other words, **when the minimum of the cardinality is zero**
  - Use a single line in ER notations

- **Mandatory** participation (Total participation)
  - One entity occurrence requires corresponding entity occurrence in particular relationship
  - In other words, **when the minimum of the cardinality is one**
  - Use a double line in ER notations

Participation Constraint (Chen Symbols)



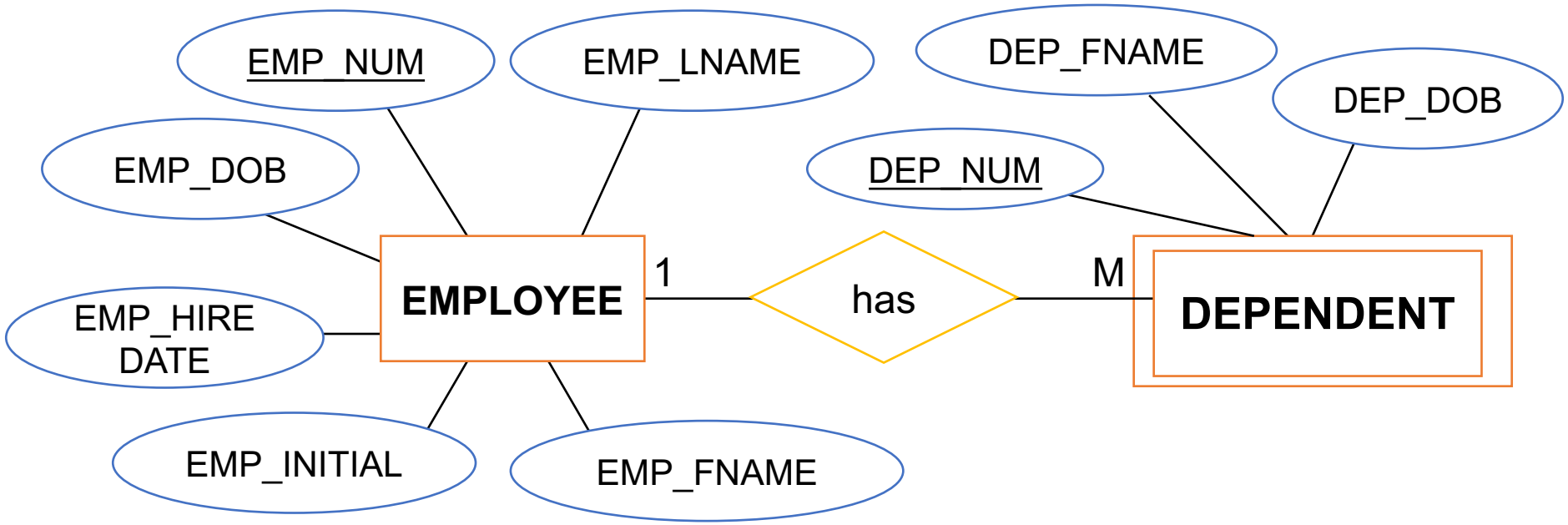
Types of Entities

- Types of Entities
  - **Strong Entities** – one that exists on its own, independent of other entities [1]
  - **Weak Entities**
    - one whose existence depends on another entity [1]
    - an entity that cannot be uniquely identified by its attributes alone [wiki]

[1] http://answers.yahoo.com/question/index?qid=20071220203606AAIhmwt

Weak Entity

- Weak entity is an entity of which existence depends on another entity.
- In other words, weak entities cannot exist if their related entities do not exist.



Weak Entity – Example (cont.)

FIGURE 4.11 A WEAK ENTITY IN A STRONG RELATIONSHIP

Table name: EMPLOYEE — Database name: Ch04_ShortCo

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_DOB	EMP_HIREDATE
1001	Callifante	Jeanine	J	12-Mar-64	25-May-97
1002	Smithson	William	K	23-Nov-70	28-May-97
1003	Washington	Herman	H	15-Aug-68	28-May-97
1004	Chen	Lydia	B	23-Mar-74	15-Oct-98
1005	Johnson	Melanie		28-Sep-66	20-Dec-98
1006	Ortega	Jorge	G	12-Jul-79	05-Jan-02
1007	O'Donnell	Peter	D	10-Jun-71	23-Jun-02
1008	Brzenski	Barbara	A	12-Feb-70	01-Nov-03

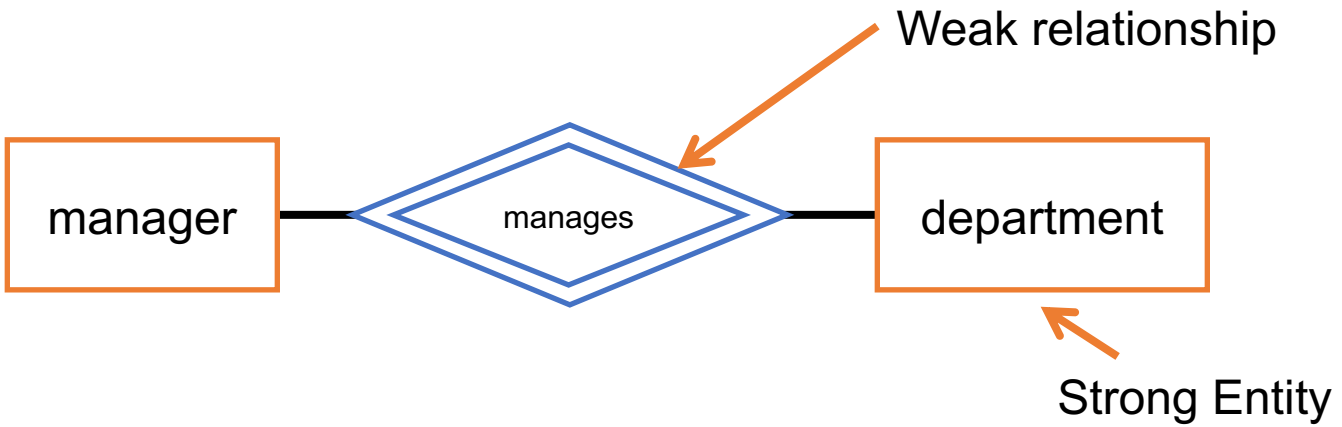
Table name: DEPENDENT

EMP_NUM	DEP_NUM	DEP_FNAME	DEP_DOB
1001	1	Annelise	05-Dec-97
1001	2	Jorge	30-Sep-02
1003	1	Suzanne	25-Jan-04
1006	1	Carlos	25-May-01
1008	1	Michael	19-Feb-95
1008	2	George	27-Jun-98
1008	3	Katherine	18-Aug-03

Types of Relationship

2 Types of Relationship

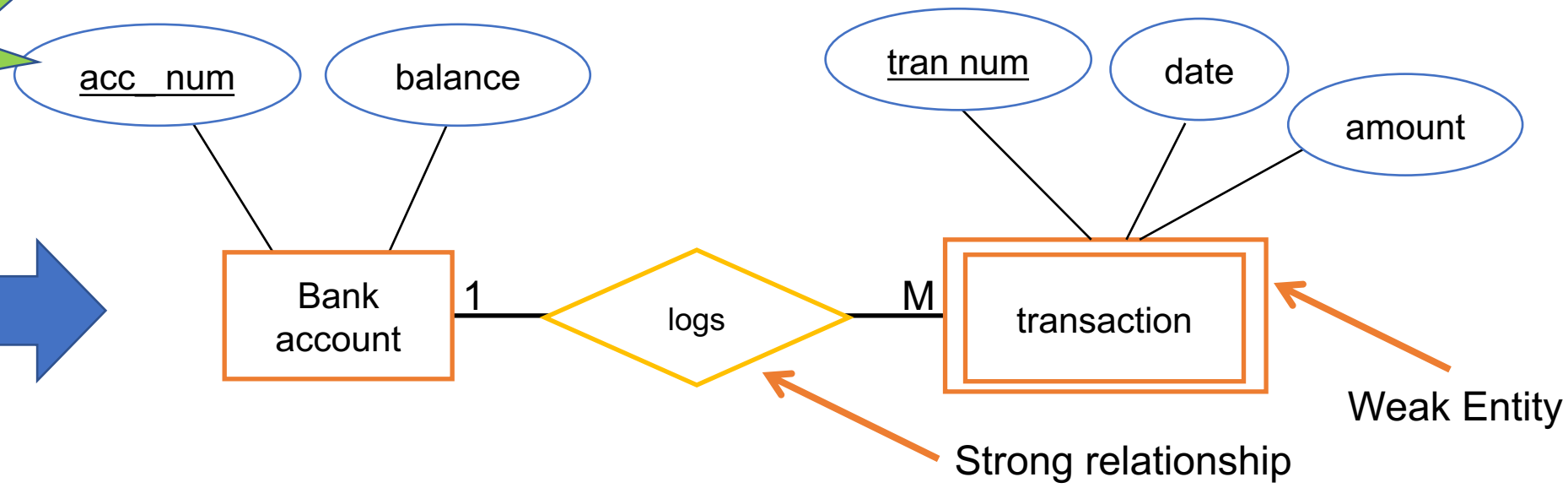
- **Weak Relationship** (Non-Identifying Relationship) – link among strong entities
- **Strong Relationship** (Identifying Relationship) – link between strong and weak entities



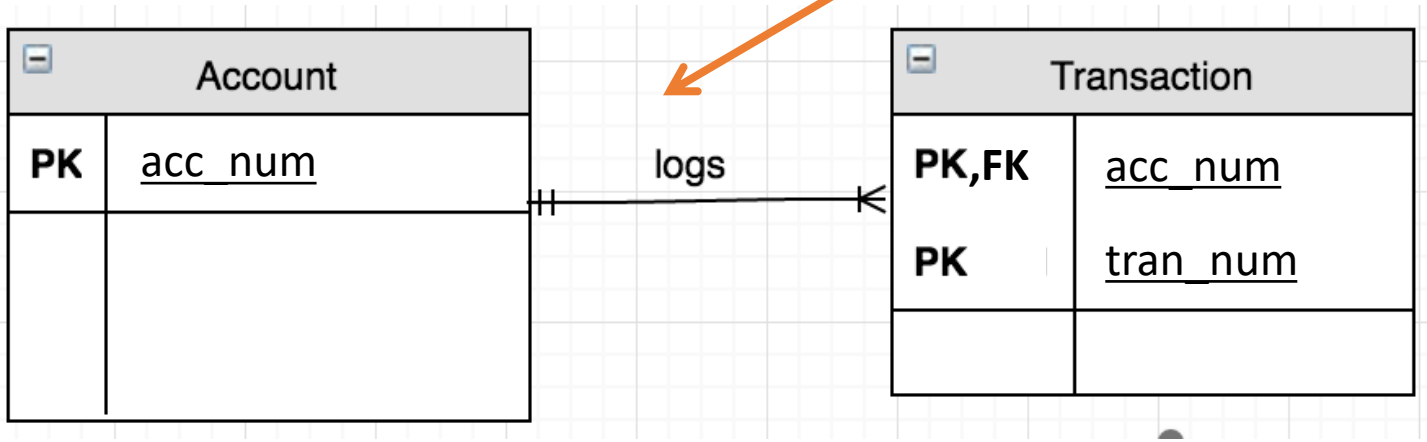
Types of Relationship – (2)

Recommend

CHEN ERD Notation must not include the Foreign Key Attribute



Only Crow Foot ERD Notation can cover the Foreign Key Attribute



Drawing ERD tools

- Microsoft Visio
  - http://holowczak.com/drawing-e-r-diagrams-with-crows-foot-notation-using-microsoft-visio-tutorial/
- https://creately.com (allow online collaboration)
- https://erdplus.com
- https://www.draw.io/

